Efficient 9-Qubit NEQR RGB Patch Encoding and Trainable Quantum Filters for Hybrid Quantum-Classical ResNet-18

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Abstract

This work overcomes the lack of full-color, multi-qubit patch encoders in quantum convolutional neural networks by introducing a 9-qubit NEQR amplitude encoder for $3 \times 3 \times 3$ RGB patches, combined with Gray-code amplitude loading and quantum-PCA compression to reduce state-preparation depth by up to 40%. We design trainable variational filters using both hardware-efficient and tree-tensor-network ansätze, integrate zero-noise extrapolation and symmetry-verification error mitigation, and replace ResNet-18's initial 7×7 convolution with parallel quantum 3×3 patch filters. Our approach aims for NISQ-feasible circuit depths, qubit overhead under 10 per patch, and benchmarks on CIFAR-100 targeting within 5% of classical ResNet-18 accuracy while demonstrating quantum resource savings.

Index Terms

quantum image representation, NEQR, RGB patch encoding, variational quantum circuits, hybrid ResNet-18

I. Background

High-quality color image classification critically depends on extracting local color features from small patches. Classical CNNs use $3 \times 3 \times 3$ RGB filters to capture spatial and color correlations, a capability missing from current quantum CNNs. In the quantum domain, FRQI encodes grayscale via rotation angles on a single qubit per pixel but loses exact amplitude retrieval [1]. NEQR stores exact q-bit intensities in basis states, requiring $n + q$ qubits for a $2^n \times 2^n$ image and scaling linearly with channels [2]. Generalized QIR (NGQR) supports arbitrary sizes by adding monitor qubits per dimension but remains single-channel and further multiplies qubit usage [3]. Bit-plane QIR (BRQI) compresses 8-bit grayscale into 4 qubits per pixel but cannot preserve arbitrary precision or color channels [19]. These limitations prevent amplitude-exact encoding of full RGB patches within NISQ qubit budgets and motivate a 9-qubit NEQR encoder with compression and trainable filters for hybrid QCNN architectures.

II. Literature Review

A. Quantum Image Representations (QIR)

Early QIR schemes focused on single-channel images. The Flexible Representation of Quantum Images (FRQI) embeds gray-scale intensities $\theta_{y,x}$ into rotation angles on a single qubit per pixel, with n position qubits for a $2^n \times 2^n$ image [1]. However, FRQI sacrifices exact amplitude retrieval and does not generalize naturally to RGB. The Novel Enhanced QIR (NEQR) encodes each q-bit pixel value exactly into computational basis states across q color qubits plus $2n$ position qubits, but its qubit cost grows linearly with channels and image dimensions [2]. To handle arbitrary sizes, the Generalized QIR (NGQR) adds 2–3 monitor qubits per dimension for non-power-of-two images, yet remains single-channel and still multiplies qubit usage per channel [3]. Bit-plane QIR (BRQI) compacts 8-bit grayscale into 4 qubits per pixel by bit-plane slicing, but cannot preserve arbitrary precision or direct color channels [19]. These limitations highlight the need for a multi-qubit patch encoder that can amplitude-exactly load full RGB patches within NISQ budgets.

B. Patch-Level Quantum Compression

To reduce qubit load per patch, quantum Principal Component Analysis (qPCA) offers exponential speedup in extracting dominant variance directions, compressing data from 27 values to $k \ll 27$ principal amplitudes in $O(k\,\mathrm{poly}(\log N))$ time [7]. Variational Quantum Autoencoders (VQAE) further demonstrate end-to-end compression and reconstruction of 3×3 RGB patches into 5–7 qubits with $> 95\%$ fidelity under shallow ansätze [8]. These patch-level methods motivate integrating compression with amplitude encoding to lower state-preparation complexity.

C. Amplitude-Loading Circuits

Loading arbitrary amplitudes into an n-qubit state naively requires $O(2^n\, n)$ gates. Gray-code tree circuits reduce that to $O(n\, 2^n)$ depth by hierarchical loading of sparse amplitudes [9]. Recent sparse state preparation optimizations eliminate gates for zero components, achieving up to 40% depth reduction when patches exhibit color redundancy [16]. Embedding these techniques into NEQR circuits will shrink depth for 27-value RGB patches.

D. Variational Filter Design and Expressivity

Quantum Convolutional Neural Networks (QCNN) adapt classical CNN concepts, using patch-based PQC filters. The original QCNN applied fixed Toffoli and CZ circuits on 10–20 qubits for many-body state classification [15]. Variational Quanvolutional Neural Networks (VQNN) introduced R_Y–CNOT layers for 2×2 grayscale patches up to 4 qubits, showing minor MNIST accuracy gains but limited expressivity [4], [5]. Effective rank metrics quantify ansatz expressivity; tree-tensor-network (TTN) ansätze aligned to patch connectivity outperform hardware-efficient layouts by capturing spatial correlations with fewer parameters [12], [18]. We will compare both ansätze to identify minimal depth achieving $> 90\%$ expressivity for color patches.

E. Error Mitigation Strategies

NISQ circuits suffer noise-induced bias. Zero-noise extrapolation runs each patch circuit at scaled noise levels $\lambda \in \{1, 2\}$ and Richardson-extrapolates $\lambda \to 0$ to recover ideal expectation values [10]. Symmetry verification discards runs violating invariants—e.g. per-channel mean of a patch amplitude—boosting effective fidelity [11]. Integrating these into patch filters is critical to obtaining reliable features on real hardware.

F. Optimizer and Training Dynamics

Hybrid QNN training faces barren plateaus with increasing qubit count [11]. Advanced optimizers like AdaBelief, Yogi, and RMSProp demonstrate smoother convergence in QCNN contexts than Adam or SGD, likely due to adaptive step sizing that counters noise-driven gradient variance [13], [14]. We will profile quantum vs. classical layer gradients, applying separate learning-rate schedules to avoid qubit-scale barren regions.

G. NISQ Resource Mapping

Mapping patch filters to current devices requires parallel patch-circuit execution. A 27-qubit device can run three 9-qubit patch circuits per cycle; a 54-qubit device runs six in parallel [4], [15]. For a 32×32 image with stride 1, $30 \times 30 = 900$ patches, processed in batches of 6 yields 150 cycles. With a per-cycle duration $\sim 200\ \mu$s, total filter time ~ 30 ms—compatible with ResNet inference latencies.

H. Identified Gaps

Despite substantial advances in quantum image representations, several critical limitations remain. First, no existing representation encodes exact amplitudes for full 3×3×3 RGB patches within nine qubits or fewer, leaving a gap in multi-channel patch encoding under tight NISQ budgets. Second, many QIR schemes suffer either excessive qubit costs or loss of precision as the number of channels grows, undermining amplitude-exact loading. Third, state-of-the-art QCNN architectures support only grayscale filters of at most four qubits, with no direct color support, preventing color feature extraction at the quantum layer. Fourth, existing variational and fixed-circuit filters lack systematic tools for trainability analysis and expressivity diagnostics, limiting performance optimization. Fifth, no prior work has integrated patch-level quantum filters into modern ResNet-style 3×3 convolution layers, leaving a disconnect between quantum encoders and classical network architectures. Finally, there is a dearth of comprehensive error-mitigation strategies and detailed resource analyses tailored to NISQ hardware requirements, which is essential for practical deployment of quantum filters.

III. Proposed Direction

A. Core Innovations

To address these gaps, we propose a sequence of core innovations. We introduce a nine-qubit NEQR encoder capable of amplitude-exact loading of 3×3×3 RGB patches by leveraging Gray-code amplitude-loading circuits, reducing state-preparation depth by up to forty percent compared with prior tree-decomposition approaches. We integrate patch-level compression via quantum principal component analysis or shallow variational quantum autoencoders, which compress twenty-seven patch amplitudes into six or seven qubits while maintaining fidelity above ninety-five percent. We design trainable variational convolutional filters employing both hardware-efficient and tree-tensor-network ansätze, systematically comparing their expressivity to identify the minimal circuit depth required to exceed ninety-percent effective rank. To mitigate noise on NISQ devices, we combine zero-noise extrapolation over two noise scales with symmetry-verification postselection on per-channel mean invariants. Finally, we embed these quantum filters into a hybrid ResNet-18 architecture by replacing the initial 7×7 convolution and pooling with parallel quantum-encoded 3×3 patch filtering, enabling direct benchmarking against classical ResNet-18 on the CIFAR-100 dataset.

IV. Methodology

A. Amplitude Encoding

Load each patch's 27 intensities v_i into 9 qubits via Gray-code circuits in $O(9\,2^9)$ gates [9]. Apply quantum PCA to compress to 6 qubits when appropriate [7].

B. Filter Ansatz

Design $L \in \{2, 3\}$ layers of:

$$U(\boldsymbol{\theta}) = \prod_{\ell=1}^{L} \big(\otimes_{q=1}^{9} R_Y(\theta_{\ell,q})\big)\big(\prod_{(q,r)\in E} \mathrm{CNOT}_{q,r}\big),$$

with E matching patch geometry [12].

C. Error Mitigation

Run patch circuits at noise scales $\lambda \in \{1, 2\}$, measure expectation $\langle Z \rangle_\lambda$, then extrapolate $\lambda \to 0$ [10]. Enforce mean-preservation symmetry per channel to postselect valid outcomes.

D. Hybrid ResNet-18

Our hybrid architecture modifies the standard ResNet-18 by replacing the initial conv$(7 \times 7, s = 2)$ plus pooling layer with a parallel quantum patch processing pipeline. The replacement procedure begins by extracting overlapping 3×3 patches from the input image with unit stride. Each extracted patch undergoes quantum encoding using our 9-qubit NEQR amplitude loading scheme, followed by variational filtering through the designed ansätze, and measurement to obtain classical feature values. The processed patches are then reassembled into $56 \times 56 \times 9$ feature maps that maintain spatial correspondence with the original convolutional output dimensions. Finally, these quantum-generated feature maps proceed through the remaining ResNet-18 blocks without modification, enabling direct comparison with the classical baseline while demonstrating the quantum enhancement at the initial feature extraction stage.

E. Training and Evaluation

Dataset: CIFAR-100 (50k train, 10k test). **Metrics**: Top-1 accuracy, convergence speed, qubit & gate counts, latency. **Baselines**: Classical ResNet-18, 4-qubit QCNN, compressed-embedding QCNN. **Optimizers**: AdaBelief, Yogi, AdaGrad, Adam, RMSProp [13], [14].

V. Expected Contributions

This research will deliver the first exact nine-qubit quantum encoder for full-color 3×3×3 RGB patches in convolutional neural networks, significantly expanding QCNN capabilities. It will demonstrate efficient patch-level compression using quantum PCA and variational quantum autoencoders, establishing a new standard for fidelity-preserving qubit reduction. By conducting a comparative expressivity study of hardware-efficient versus tree-tensor-network variational filters, the work will provide actionable guidelines for ansatz selection in quantum image processing. The integration of zero-noise extrapolation and symmetry verification into the quantum filter pipeline will advance error-mitigation practices tailored to variational circuits. Finally, the hybrid quantum-classical ResNet-18 implementation and CIFAR-100 benchmarks will empirically validate NISQ-feasible accuracy within five percent of the classical baseline, demonstrating tangible quantum resource savings.

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